

Initiating Search

November 3, 2025, 9:12 PM

• Search:

CAS Lexicon Search:

Heck reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (8,455)	 References	View Results
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Filtered By:		
Document Type:	Journal	
Language:	English	
Publication Name:	Organic Process Research & Development, Journal of the American Chemical Society, Angewandte Chemie, International Edition	

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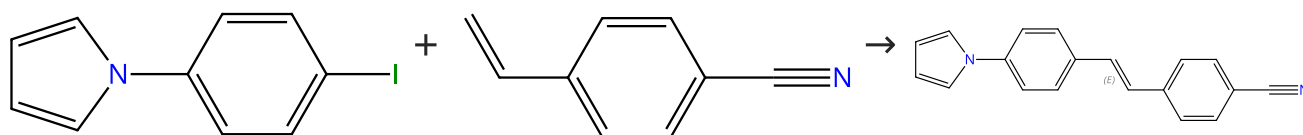
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Reactions (30)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (54)

 Suppliers (60)

Double bond geometry shown

31-177-CAS-23309358

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Sodium chloride, Tripotassium phosphate, Poly (oxy-1,2-ethanediyl), α -[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]- ω -methoxy-
Catalysts: Iron, Palladium
Solvents: Dimethylformamide, Water; 16 h, 45 °C

Experimental Protocols

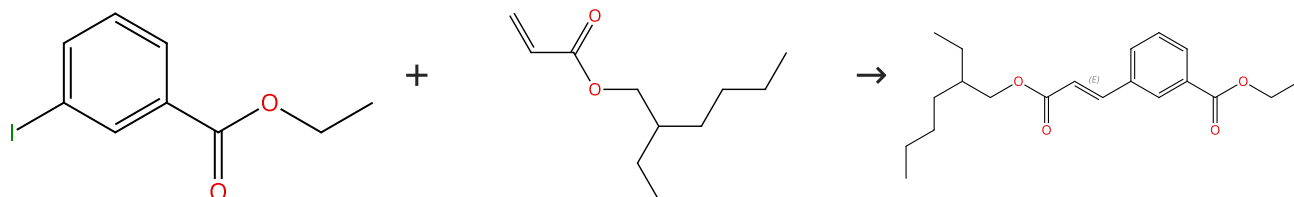
Water-Sculpting of a Heterogeneous Nanoparticle Precatalyst for Mizoroki-Heck Couplings under Aqueous Micellar Catalysis Conditions

By: Pang, Haobo; et al

Journal of the American Chemical Society (2021), 143(9), 3373-3382.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (76)

 Suppliers (64)

Double bond geometry shown

31-177-CAS-23308906

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Sodium chloride, Tripotassium phosphate, Poly (oxy-1,2-ethanediyl), α -[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]- ω -methoxy-
Catalysts: Iron, Palladium
Solvents: Dimethylformamide, Water; 16 h, 45 °C

Experimental Protocols

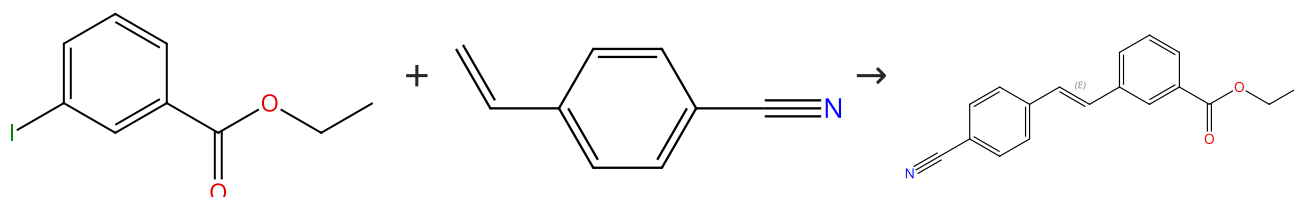
Water-Sculpting of a Heterogeneous Nanoparticle Precatalyst for Mizoroki-Heck Couplings under Aqueous Micellar Catalysis Conditions

By: Pang, Haobo; et al

Journal of the American Chemical Society (2021), 143(9), 3373-3382.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (76)

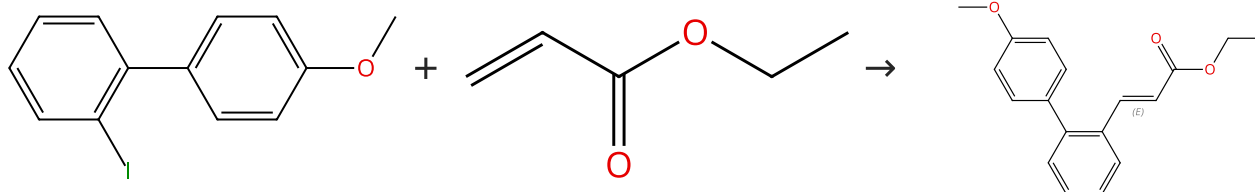
 Suppliers (60)

Double bond geometry shown

31-177-CAS-23309802 Steps: 1 Yield: 100% 1.1 Reagents: Sodium chloride, Tripotassium phosphate, Poly (oxy-1,2-ethanediyl), α-[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2<i>H</i>-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]-ω-methoxy- Catalysts: Iron, Palladium Solvents: Dimethylformamide, Water; 16 h, 45 °C Experimental Protocols	Water-Sculpting of a Heterogeneous Nanoparticle Precatalyst for Mizoroki-Heck Couplings under Aqueous Micellar Catalysis Conditions By: Pang, Haobo; et al Journal of the American Chemical Society (2021), 143(9), 3373-3382.
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Scheme 4 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (8)

Suppliers (76)

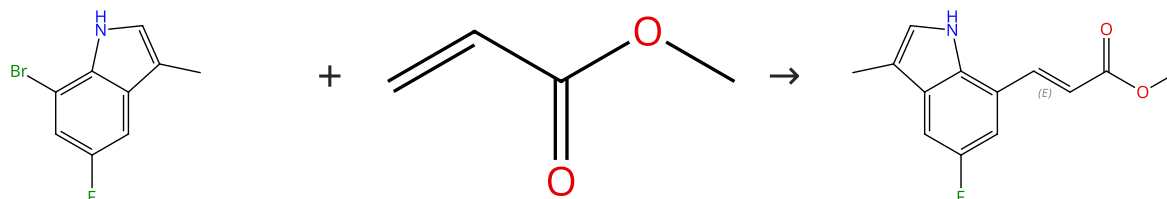
Double bond geometry shown

Suppliers (7)

31-177-CAS-4232227 Steps: 1 Yield: 100% 1.1 Reagents: Sodium bicarbonate, Tetrabutylammonium chloride Catalysts: Palladium diacetate Solvents: Dimethylformamide; 2 d, 100 °C; 100 °C → rt Experimental Protocols	Aryl to Aryl Palladium Migration in the Heck and Suzuki Coupling of o-Halobiaryls By: Campo, Marino A.; et al Journal of the American Chemical Society (2007), 129(19), 6298-6307.
31-177-CAS-7309060 Steps: 1 Yield: 100% 1.1 Reagents: Sodium bicarbonate, Tetrabutylammonium chloride Catalysts: Palladium diacetate Solvents: Dimethylformamide; 1 d, 100 °C Experimental Protocols	Novel 1,4-Palladium Migration in Organopalladium Intermediates Derived from o-Iodobiaryls By: Campo, Marino A.; et al Journal of the American Chemical Society (2002), 124(48), 14326-14327.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (43)

Suppliers (60)

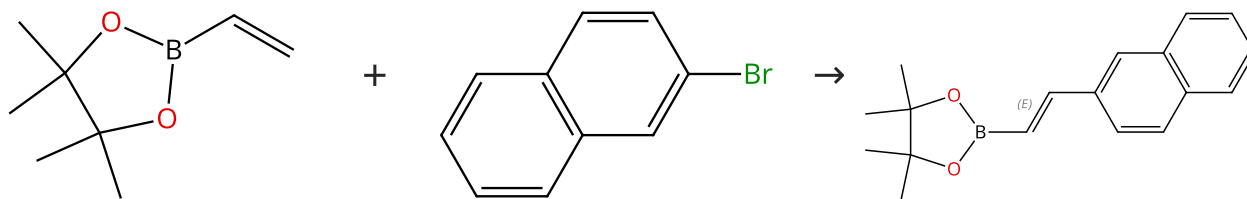
Double bond geometry shown

Suppliers (3)

31-177-CAS-5104864 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine Catalysts: Palladium diacetate, Tri-<i>o</i>-tolylphosphine Solvents: Acetonitrile; rt; rt → reflux; overnight, reflux; reflux → rt Experimental Protocols	Development of a Scalable Process for DG-041, a Potent EP₃ Receptor Antagonist, via Tandem Heck Reactions By: Zegar, Siead; et al Organic Process Research & Development (2007), 11(4), 747-753.
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Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

Suppliers (87)

Double bond geometry shown

Suppliers (7)

31-177-CAS-18902503

Steps: 1 Yield: 100%

Interrogating Pd(II) Anion Metathesis Using a Bifunctional Chemical Probe: A Transmetalation Switch

By: Molloy, John J.; et al

Journal of the American Chemical Society (2018), 140(1), 126-130.

1.1 Reagents: Triethylamine

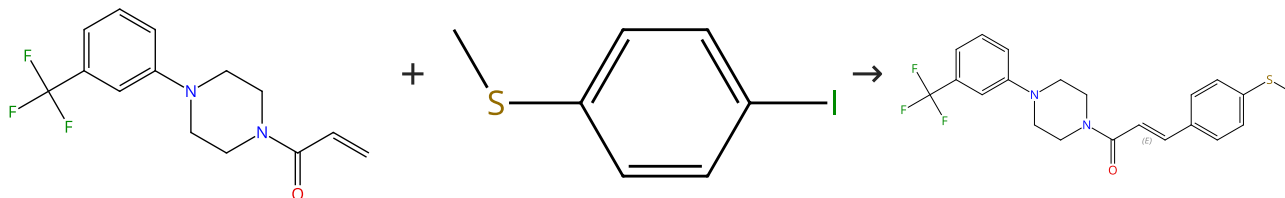
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl

Solvents: 1,4-Dioxane; 24 h, 80 °C

Experimental Protocols

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (64)

Double bond geometry shown

31-177-CAS-23335608

Steps: 1 Yield: 100%

Water-Sculpting of a Heterogeneous Nanoparticle Precatalyst for Mizoroki-Heck Couplings under Aqueous Micellar Catalysis Conditions

By: Pang, Haobo; et al

Journal of the American Chemical Society (2021), 143(9), 3373-3382.

1.1 Reagents: Sodium chloride, Tripotassium phosphate, Poly(oxy-1,2-ethanediyl), α-[4-[[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]-ω-methoxy-

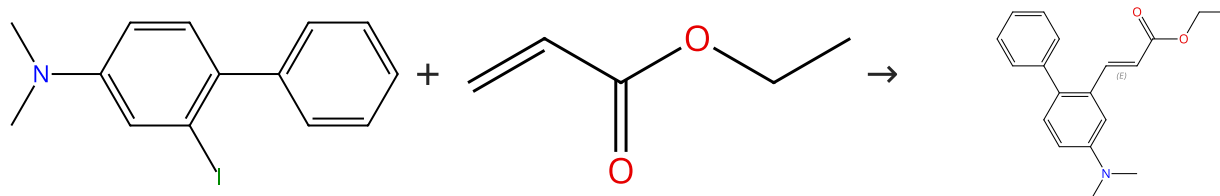
Catalysts: Iron, Palladium

Solvents: Dimethylformamide, Water; 16 h, 45 °C

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (76)

Double bond geometry shown

31-177-CAS-15210320

Steps: 1 Yield: 100%

Aryl to Aryl Palladium Migration in the Heck and Suzuki Coupling of o-Halobiaryls

By: Campo, Marino A.; et al

Journal of the American Chemical Society (2007), 129(19), 6298-6307.

1.1 Reagents: Sodium bicarbonate, Tetrabutylammonium chloride

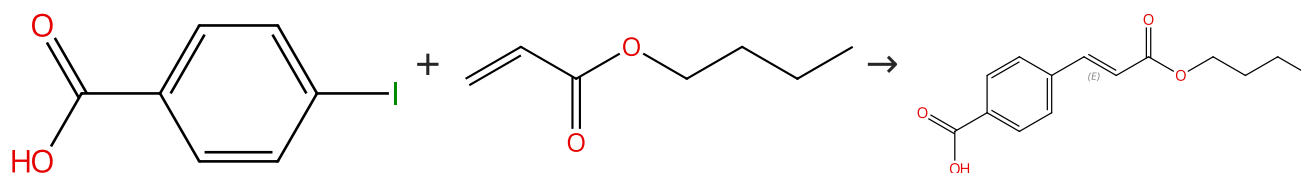
Catalysts: Palladium diacetate

Solvents: Dimethylformamide; 1 d, 100 °C; 100 °C → rt

Experimental Protocols

Scheme 9 (2 Reactions)

Steps: 1 Yield: 100%



resin-bound

Suppliers (64)

Double bond geometry shown

Supplier (1)

31-177-CAS-3434833

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Sodium acetate
Catalysts: Tetrabutylammonium bromide, Di- μ -chlorobis[2-chloro-3-(dimethylamino- κ M)-1-phenyl-1-propen-1-yl- κ C] dipalladium
Solvents: Dimethylacetamide; 130 °C
- 1.2 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; rt

Experimental Protocols

Kinetics and Mechanistic Aspects of the Heck Reaction Promoted by a CN-Palladacycle

By: Consorti, Crestina S.; et al

Journal of the American Chemical Society (2005), 127(34), 12054-12065.

31-177-CAS-7712824

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Sodium acetate
Catalysts: Tetrabutylammonium bromide, Di- μ -chlorobis[2-chloro-3-(dimethylamino- κ M)-1-phenyl-1-propen-1-yl- κ C] dipalladium
Solvents: Dimethylacetamide; 15 min, 30 °C
- 1.2 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; rt

Experimental Protocols

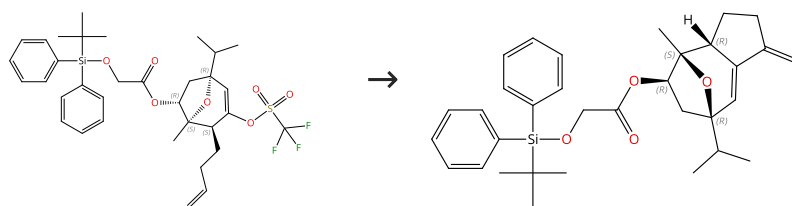
Kinetics and Mechanistic Aspects of the Heck Reaction Promoted by a CN-Palladacycle

By: Consorti, Crestina S.; et al

Journal of the American Chemical Society (2005), 127(34), 12054-12065.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (+)Absolute stereochemistry shown,
Rotation (-)

31-177-CAS-20140891

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Triethylamine
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Toluene; 30 min, 80 °C
- 1.2 **Reagents:** Water

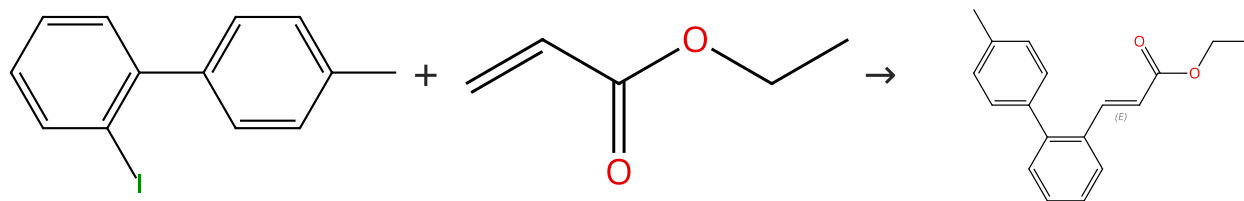
 β -Ketoesters as Mono- or Bisnucleophiles: A Concise Enantioselective Total Synthesis of (-)-Englerin A and B

By: Guo, Lei; et al

Angewandte Chemie, International Edition (2019), 58(25), 8346-8350.

Scheme 11 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (37)

Suppliers (76)

Double bond geometry shown

31-177-CAS-15492417

Steps: 1 Yield: 100%

Aryl to Aryl Palladium Migration in the Heck and Suzuki Coupling of o-Halobiaryls

By: Campo, Marino A.; et al

Journal of the American Chemical Society (2007), 129(19), 6298-6307.

1.1 **Reagents:** Sodium bicarbonate, Tetrabutylammonium chloride
Catalysts: Palladium diacetate
Solvents: Dimethylformamide; 1 d, 100 °C; 100 °C → rt

Experimental Protocols

31-177-CAS-7665350

Steps: 1 Yield: 100%

Novel 1,4-Palladium Migration in Organopalladium Intermediates Derived from o-Iodobiaryls

By: Campo, Marino A.; et al

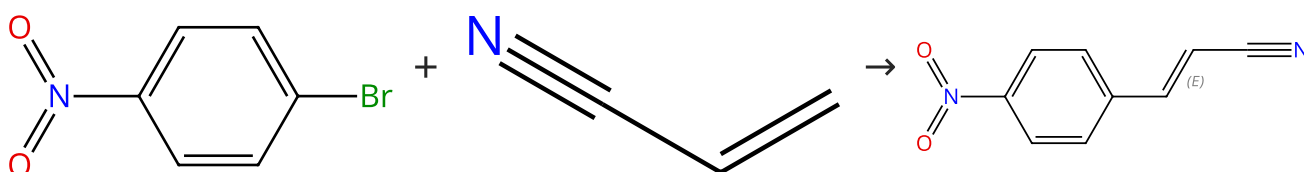
Journal of the American Chemical Society (2002), 124(48), 14326-14327.

1.1 **Reagents:** Sodium bicarbonate, Tetrabutylammonium chloride
Catalysts: Palladium diacetate
Solvents: Dimethylformamide; 1 d, 100 °C

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (68)

Double bond geometry shown

Suppliers (36)

31-177-CAS-1358797

Steps: 1 Yield: 100%

Ligandless Heck Coupling between a Halogenated Aniline and Acrylonitrile Catalyzed by Pd/C: Development and Optimization of an Industrial-Scale Heck Process for the Production of a Pharmaceutical Intermediate

By: Schils, Didier; et al

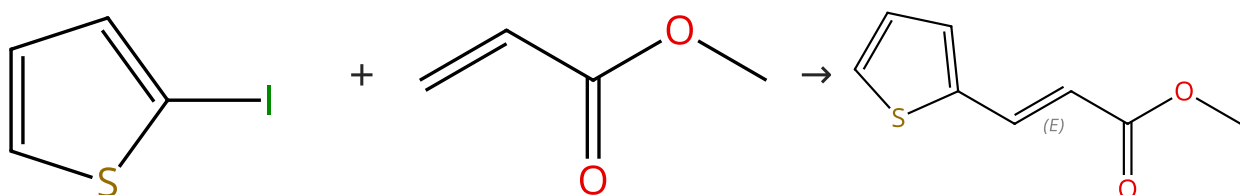
Organic Process Research & Development (2008), 12(3), 530-536.

1.1 **Reagents:** Sodium acetate
Catalysts: Palladium
Solvents: Dimethylacetamide; 20 h, 140 °C

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (78)

Suppliers (60)

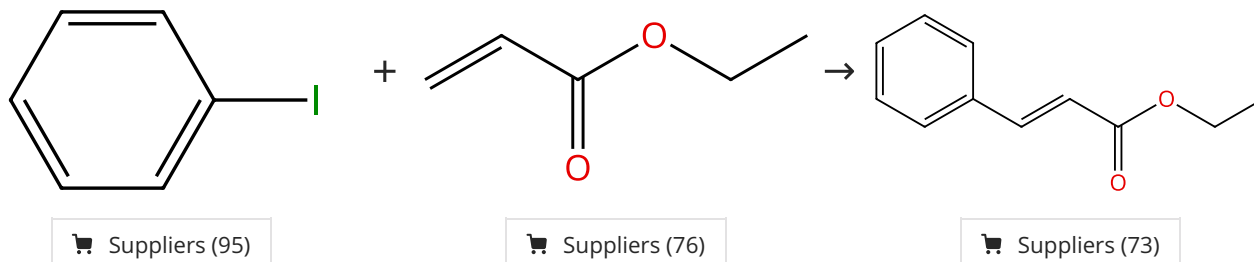
Double bond geometry shown

Suppliers (34)

31-177-CAS-1509429 Steps: 1 Yield: 100%	Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts By: Pandarus, Valerica; et al Organic Process Research & Development (2012), 16(1), 117-122.
1.1 Reagents: Sodium acetate Catalysts: Palladium (silica-supported) Solvents: Dimethylformamide; 20 h, 90 - 120 °C	
Experimental Protocols	

Scheme 14 (1 Reaction)

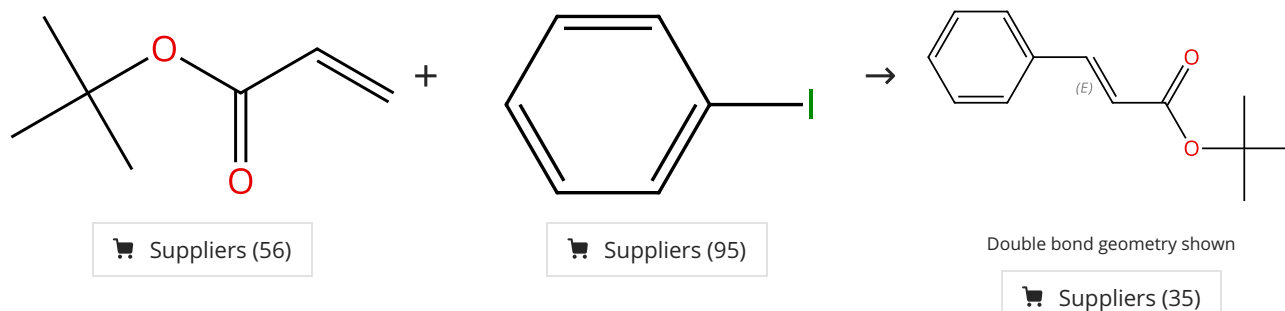
Steps: 1 Yield: 100%



31-614-CAS-27642022 Steps: 1 Yield: 100%	Preparation of Recyclable and Versatile Porous Poly(aryl thioether)s by Reversible Pd-Catalyzed C-S/C-S Metathesis By: Rivero-Crespo, Miguel A.; et al Journal of the American Chemical Society (2021), 143(50), 21331-21339.
1.1 Reagents: Diisopropylethylamine Catalysts: Palladium chloride, 1,1':3',1''-Terphenyl, 4,4''-bis (phenylthio)-5'-[4-(phenylthio)phenyl]-, homopolymer Solvents: Toluene; 18 h, 100 °C	
Experimental Protocols	

Scheme 15 (1 Reaction)

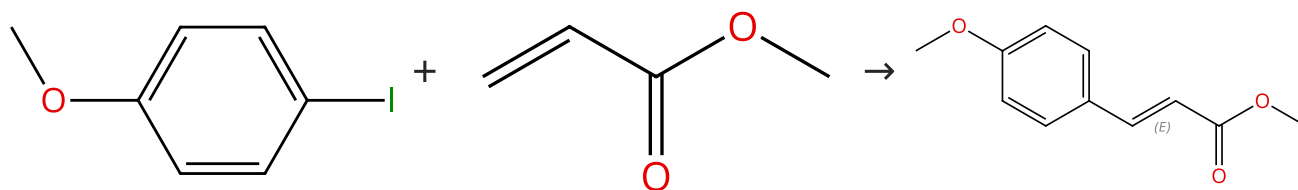
Steps: 1 Yield: 100%



31-177-CAS-8554492 Steps: 1 Yield: 100%	Palladium-Catalyzed C-C Coupling under Thermomorphic Conditions By: Bergbreiter, David E.; et al Journal of the American Chemical Society (2000), 122(38), 9058-9064.
1.1 Reagents: Triethylamine Solvents: Pentane 1.2 Reagents: Palladium (complexes with phosphine-functionalized poly(N-isopropylacrylamide)) Solvents: Ethanol	
Experimental Protocols	

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (88)

Suppliers (60)

Double bond geometry shown

Suppliers (49)

31-177-CAS-10912793

Steps: 1 Yield: 100%

Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts

By: Pandarus, Valerica; et al

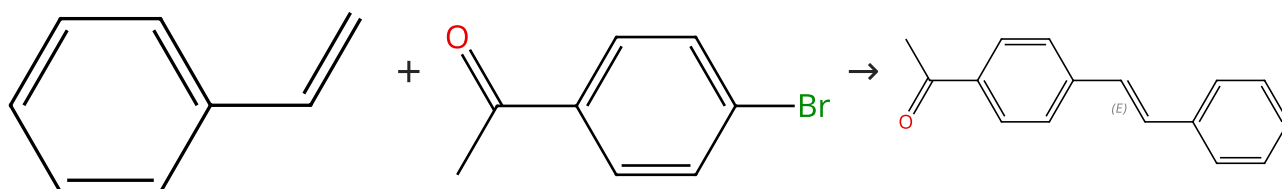
Organic Process Research & Development (2012), 16(1), 117-122.

1.1 **Reagents:** Sodium acetate
Catalysts: Palladium (silica-supported)
Solvents: Dimethylformamide; 20 h, 90 - 120 °C

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (115)

Suppliers (84)

Double bond geometry shown

Suppliers (11)

31-177-CAS-6370481

Steps: 1 Yield: 100%

Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts

By: Pandarus, Valerica; et al

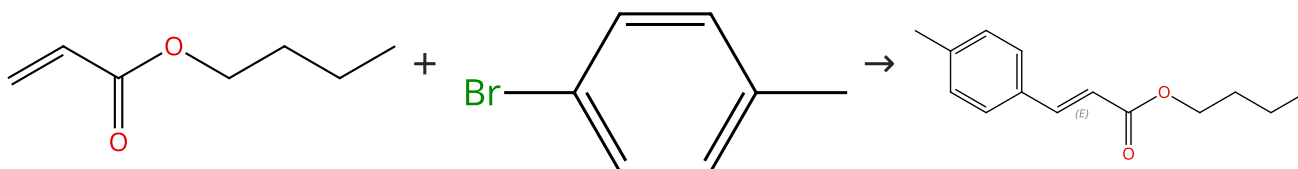
Organic Process Research & Development (2012), 16(1), 117-122.

1.1 **Reagents:** Sodium acetate
Catalysts: Palladium (silica-supported)
Solvents: Dimethylformamide; 20 h, 90 - 120 °C

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (64)

Suppliers (67)

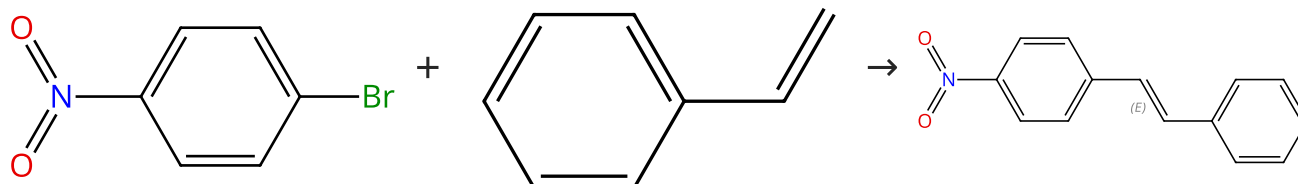
Double bond geometry shown

Suppliers (10)

31-177-CAS-8355248	Steps: 1 Yield: 100%	A Fluorescence-Based Assay for High-Throughput Screening of Coupling Reactions. Application to Heck Chemistry
1.1 Reagents: Sodium acetate Catalysts: Bis(dibenzylideneacetone)palladium, 1,1'-Bis(di- <i>tert</i> -butylphosphino)ferrocene Solvents: Dimethylformamide; 2 h, 75 °C		By: Shaughnessy, Kevin H.; et al Journal of the American Chemical Society (1999), 121(10), 2123-2132.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (115)

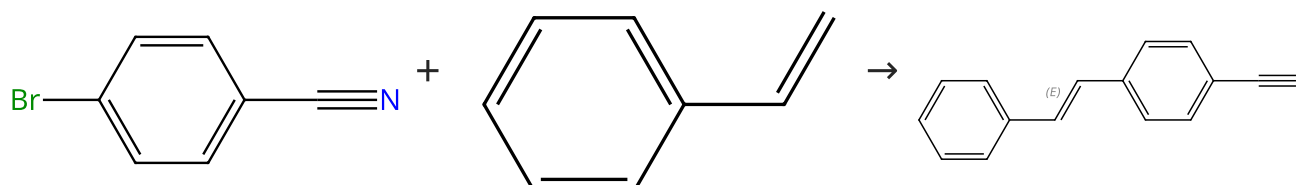
Double bond geometry shown

Suppliers (56)

31-177-CAS-1829998	Steps: 1 Yield: 100%	Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts
1.1 Reagents: Sodium acetate Catalysts: Palladium (silica-supported) Solvents: Dimethylformamide; 20 h, 90 - 120 °C		By: Pandarus, Valerica; et al Organic Process Research & Development (2012), 16(1), 117-122.
Experimental Protocols		

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (92)

Suppliers (115)

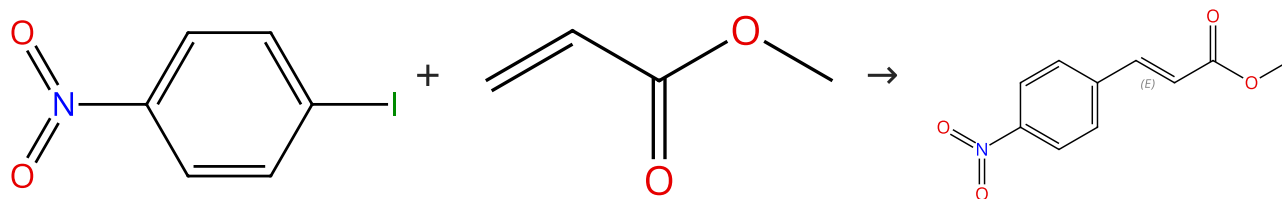
Double bond geometry shown

Suppliers (56)

31-177-CAS-10631497	Steps: 1 Yield: 100%	Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts
1.1 Reagents: Sodium acetate Catalysts: Palladium (silica-supported) Solvents: Dimethylformamide; 20 h, 90 °C		By: Pandarus, Valerica; et al Organic Process Research & Development (2012), 16(1), 117-122.
Experimental Protocols		

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

Suppliers (60)

Double bond geometry shown

Suppliers (57)

31-177-CAS-12757804

Steps: 1 Yield: 100%

Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts

By: Pandarus, Valerica; et al

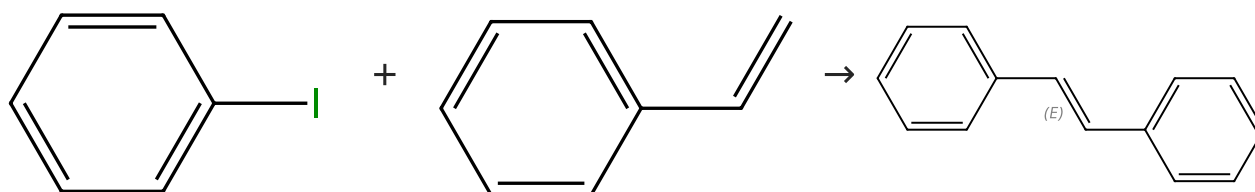
Organic Process Research & Development (2012), 16(1), 117-122.

1.1 Reagents: Sodium acetate
Catalysts: Palladium (silica-supported)
Solvents: Dimethylformamide; 20 h, 90 °C

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (95)

Suppliers (115)

Double bond geometry shown

Suppliers (78)

31-177-CAS-8464205

Steps: 1 Yield: 100%

High Catalytic Activity of Palladium(II)-Exchanged Mesoporous Sodalite and NaA Zeolite for Bulky Aryl Coupling Reactions: Reusability under Aerobic Conditions

By: Choi, Minkee; et al

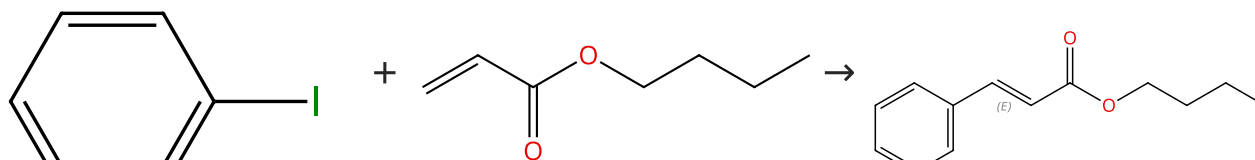
Angewandte Chemie, International Edition (2009), 48(20), 3673-3676, S3673/1-S3673/5.

1.1 Catalysts: Palladium nitrate
Solvents: Dimethylformamide, Water; 3 h, 100 °C

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (95)

Suppliers (64)

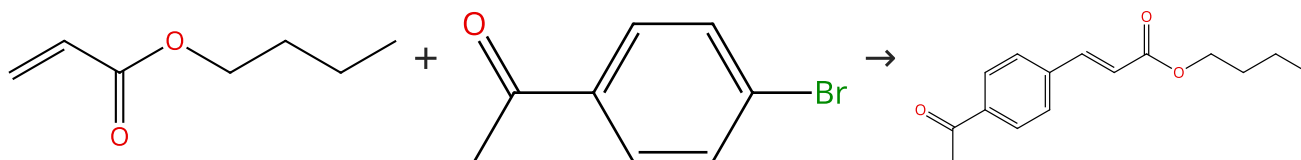
Double bond geometry shown

Suppliers (16)

31-177-CAS-14384513 Steps: 1 Yield: 100% 1.1 Reagents: Sodium acetate Catalysts: Tetrabutylammonium bromide, Di-μ-chlorobis[2-chloro-3-(dimethylamino-κM)-1-phenyl-1-propen-1-yl-κC] dipalladium Solvents: Dimethylacetamide; 15 min, 30 °C Experimental Protocols	Kinetics and Mechanistic Aspects of the Heck Reaction Promoted by a CN-Palladacycle By: Consorti, Crestina S.; et al Journal of the American Chemical Society (2005), 127(34), 12054-12065.
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Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (64)

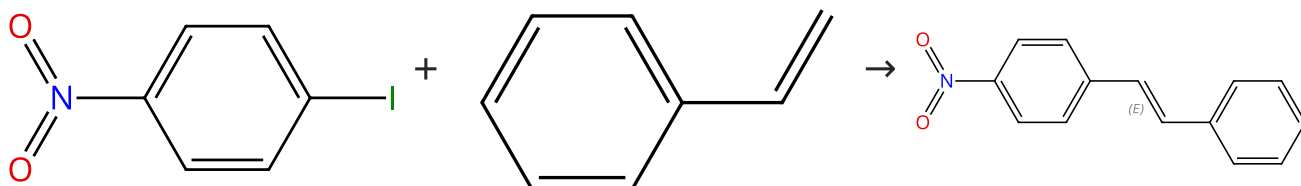
Suppliers (84)

Supplier (1)

31-177-CAS-6409268 Steps: 1 Yield: 100% 1.1 Reagents: Sodium acetate Catalysts: Palladium, dichloro-2,4,6-cycloheptatrien-1-ylidene (triphenylphosphine)-, (<i>SP</i>-4-3)-, compd. with acetonitrile (1:1) Solvents: Dimethylacetamide; rt → 145 °C; 14 h, 145 °C	A carbocyclic carbene as an efficient catalyst ligand for C-C coupling reactions By: Herrmann, Wolfgang A.; et al Angewandte Chemie, International Edition (2006), 45(23), 3859-3862.
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Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

Suppliers (115)

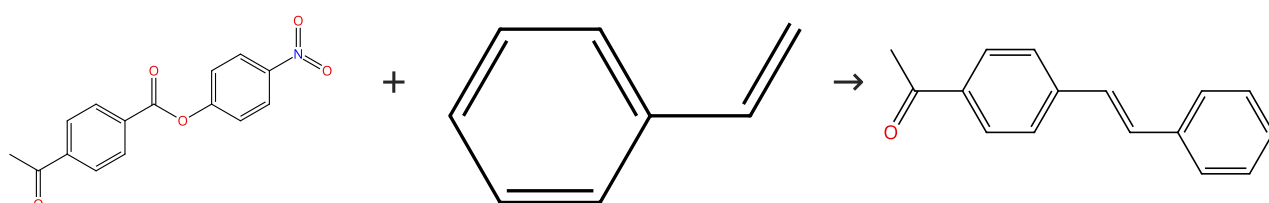
Double bond geometry shown

Suppliers (56)

31-177-CAS-15219373 Steps: 1 Yield: 100% 1.1 Reagents: Sodium acetate Catalysts: Palladium (silica-supported) Solvents: Dimethylformamide; 20 h, 90 - 120 °C Experimental Protocols	Efficient Screening and Library Generation in Parallel C-C Coupling Reactions Mediated by Organosilica SiliaCat Palladium Catalysts By: Pandarus, Valerica; et al Organic Process Research & Development (2012), 16(1), 117-122.
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Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

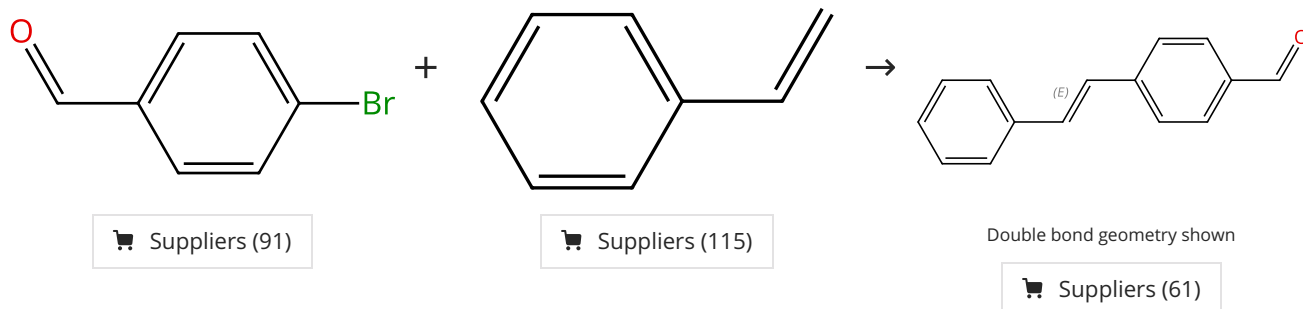
Suppliers (115)

Suppliers (3)

31-177-CAS-18106946 Steps: 1 Yield: 99% 1.1 Catalysts: Isoquinoline, Lithium chloride, Palladium chloride Solvents: <i>N</i>-Methyl-2-pyrrolidone	Pd-catalyzed decarbonylative olefination of aryl esters: towards a waste-free Heck reaction By: Goossen, Lukas J.; et al Angewandte Chemie, International Edition (2002), 41(7), 1237-1241.
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Scheme 27 (1 Reaction)

Steps: 1 Yield: 99%



31-177-CAS-4149278 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Zinc, [1-[4-(carboxy-κO)-2-methylphenyl]-3-[[1-(4-carboxy-2-methylphenyl)-1<i>H</i>-imidazolium-3-yl]methyl]-1<i>H</i>-imidazoliumato(2-)]dichloro-, hydrate (1:8) (r.p. with palladium acetate) Solvents: Dimethylacetamide; 1 h, 100 °C Experimental Protocols	Assembly and Post-Modification of a Metal-Organic Nanotube for Highly Efficient Catalysis By: Kong, Guo-Qiang; et al Journal of the American Chemical Society (2012), 134(48), 19851-19857.
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